

A Liquid Sound Retrieval using History of Velocities in Physically-based Simulation

Hyuga Saito
University of Toyama
JAPAN
m2171121@ems.u-toyama.ac.jp

Syuhei Sato
University of Toyama
JAPAN
ssato@eng.u-toyama.ac.jp

Yoshinori Dobashi
Hokkaido University
JAPAN
doba@ime.ist.hokudai.ac.jp

ABSTRACT

This paper presents a novel method for synthesizing sound effects for fluid animation. Previous approaches synthesize the sound for fluids by physically-based simulation, but these approaches need a huge computational cost. To address this, we propose a data-driven method for synthesizing sound effects for fluids. In this paper, we focus on liquid sound. A liquid sound database which consists of a set of recorded sound clips is prepared in advance, and then the most suitable sound clip for an input liquid motion is automatically retrieved from the database. The retrieval is achieved by comparing a waveform of the sound and a history of velocity of the liquid motion computed by the simulation. The velocity history is computed for the regions where the liquid sound is expected to occur. Then, a distance between the velocity history and the waveform of each sound clip in the database is calculated, and our system chooses the clip with the minimum distance. Our method achieves fast synthesis of liquid sound for simulated liquid motion, once the database is prepared. In this paper, we use a small database containing a few sound clips and evaluate the effectiveness of our retrieval approach, as a preliminary experiment of this research.

CCS CONCEPTS

• **Computing methodologies** → **Animation**; *Physical simulation*.

KEYWORDS

liquid simulation, synthesizing liquid sound, database, dynamic time warping

ACM Reference Format:

Hyuga Saito, Syuhei Sato, and Yoshinori Dobashi. 2021. A Liquid Sound Retrieval using History of Velocities in Physically-based Simulation. In *SIGGRAPH Asia 2021 Posters (SA '21 Posters)*, December 14–17, 2021, Tokyo, Japan. ACM, New York, NY, USA, 2 pages. <https://doi.org/10.1145/3476124.3488643>

1 INTRODUCTION

Physically-based fluid simulation is widely used for entertainment applications such as video games and movies. Realistic animations

can be created using the fluid simulation; these enhance the quality of various synthetic scenes in those entertainment applications. However, physics simulation need huge computational costs to create realistic animations, so different approaches have been proposed to achieve fast computation for fluid simulation.

Fluid sound is also an important factor to create realistic animations. The fluid sound can be created via various approaches such as, recording real sound, physics-based approaches, and procedural approaches. Obviously, the recording of real sound can produce realistic sound effects, but reproducing the same situation in the real world as those of the desired virtual scene is often difficult. A physics-based sound synthesis can create different sound effects and the results are relatively realistic [Langlois et al. 2016; Zheng and James 2009]. These approaches calculate vibrations of liquid surfaces arising from motions of bubbles using highly-accurate liquid simulation, and the sound is synthesized based on the vibrations. However, their computational costs are often huge; a day might be taken.

To achieve a fast synthesis of the fluid sound, we employ a data-driven approach using a liquid sound database and propose a retrieval method to obtain a sound clip from the database that suites a given liquid motion computed by fluid simulation. Our approach compares a history of velocities by the liquid simulation with a waveform of liquid sound. Our idea of using the history of velocities is based on "the fact that the liquid sound is generated from bubbles" and "an assumption that the number and sizes of the bubbles are linearly proportional to the liquid velocities". We calculate the history at regions where the liquid sound is expected to occur. Such regions are detected using a height field of the liquid surface. Then, a distance between the history of velocities and a waveform of the sound is calculated for each sound clip in the database, and our system chooses the clip with the minimum distance. The distance is calculated using Dynamic Time Warping (DTW).

Our method achieves faster sound synthesis once the liquid sound database is prepared. In this paper, we use a small database containing a few sound clips and evaluate the ability of our method to choose the optimal sound clip from the database for a simulated liquid motion.

2 THE METHOD

We simulate motions of liquid by using FLIP (Fluid-Implicit-Particle) and classify the liquid particles into two types: particles related to sound sources and others. We call these particles as source particles and non-source particles, respectively. We assume that the source particles are in splashes or falling water masses. The source particles are detected by using a height field generated from the

Permission to make digital or hard copies of part or all of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).
SA '21 Posters, December 14–17, 2021, Tokyo, Japan
© 2021 Copyright held by the owner/author(s).
ACM ISBN 978-1-4503-8687-6/21/12.
<https://doi.org/10.1145/3476124.3488643>

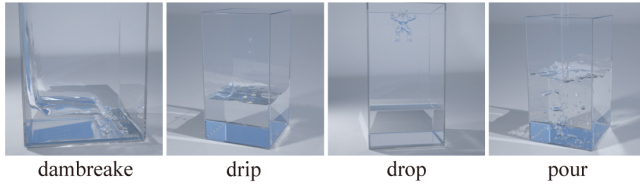


Figure 1: Snapshots of liquid simulations used for calculating the sound data in Langlois's method [Langlois et al. 2016].

liquid surface. The height field is represented with a 2D grid and the height of the highest particles is stored at each grid point. We then apply the Fourier transform to the height field and extract its low-frequency components to obtain a smooth height field. We identify the particles as the source particles if their heights are higher than the smoothed version of the height field. Additionally, the particles artificially added during simulation, such as the ones for falling liquid masses or pouring water from a faucet, are also identified as source particles.

Next, we detect the source particles colliding with the liquid surface, which we call colliding particles. To detect the colliding particles, a 3D grid is prepared and the type of the nearest particle (either source or non-source) is stored at each grid point. A source particle moving into the region of the non-source particles changes its status to a colliding particle. After detecting all the colliding particles, we compute the maximum velocity among them and store its norm for each frame. The history of the norms of the maximum velocities throughout the animation is used for the sound retrieval from the database.

The sound retrieval is based on the distance between the waveform of each sound clip in the database and the history of the velocity computed above. We use Dynamic Time Warping (DTW) [Keogh and Ratanamahatana 2005] to compute the distance between the history and the waveform, and the sound clip with the minimum distance is presented to the user. The user then manually tweaks the retrieved sound clip to match the input liquid animation to create the final sound effect.

3 EXPERIMENTAL RESULTS

This section shows some results created using our method. We use four liquid sound clips created by Langlois's physically-based method [Langlois et al. 2016] for our sound database. Each clip is named dambreak, drip, drop, pour (see Fig. 1). Fig. 2 shows input liquid animations calculated using MantaFlow [Thuerey and Pfaff 2018]. In this paper, for evaluating effectiveness of our retrieval, we created these inputs such that Fig. 2 (a), (b), (c) are similar to dambreak, drip, drop, respectively. Fig. 3 shows the history of velocities (blue curves) and the waveform of the sound clip (orange curves), where the horizontal axis indicates time in second and the vertical axis is an amplitude (the waveform) or a norm of the velocity (the history), where the amplitudes and velocities are normalized.

As shown in Fig. 3, the shape of the history of velocities is similar to the corresponding waveform. In addition, the distance between the sound clip and the velocity history computed under similar situations was minimum for all the input liquid simulations. The

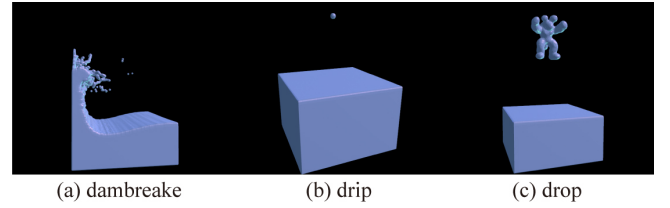


Figure 2: Snapshots of input liquid simulations.

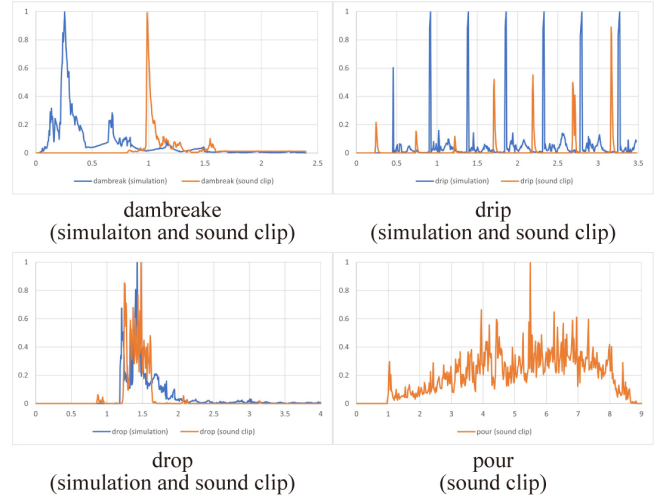


Figure 3: Graphs for the history of velocities and the waveform of the sound data.

computation time for our method was less 10% of it for the liquid simulation. The videos corresponding to these experiments can be found in the supplementary material.

4 CONCLUSIONS

We have proposed the method for synthesizing sound effects for the liquid animation using the sound database. Our system retrieved a sound clip to suite a given liquid animation from the database.

Currently, our system only retrieves the most similar sound clip to a given liquid animation, so users must tweak the sound clip manually to suite the animation. We therefore are planning to develop a method for automating this manual adjustment process.

ACKNOWLEDGMENTS

This work was supported by JSPS KAKENHI Grant Numbers JP20K19945 and JP20H05954.

REFERENCES

- E. Keogh and C. Ratanamahatana. 2005. Exact indexing of dynamic time warping. *Knowledge and Information Systems* 7 (2005), 358–386.
- Timothy R. Langlois, Changxi Zheng, and Doug L. James. 2016. Toward Animating Water with Complex Acoustic Bubbles. *ACM Trans. Graph.* 35, 4, Article 95 (2016).
- Nils Thuerey and Tobias Pfaff. 2018. MantaFlow. <http://mantaflow.com>.
- Changxi Zheng and Doug L. James. 2009. Harmonic fluids. *ACM Trans. Graph.* 28, 3, Article 37 (2009).